INCH-POUND

MIL-C-3066C(AR)
23 January 1995
SUPERSEDING
MIL-C-3066B(AR)
28 February 1969

MILITARY SPECIFICATION

CARTRIDGES, CALIBER .50: ARMOR-PIERCING-INCENDIARY, M8; AND ARMOR-PIERCING-INCENDIARY-TRACER, M20

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for Cartridges, Caliber .50: Armor-Piercing-Incendiary, M8 and Armor-Piercing-Incendiary-Tracer, M20 for use in caliber .50 weapons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC** 1305

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SPECIFICATIONS

MILITARY

MIL-S-13812	-	Steel Plate, Wrought Homogeneous for Ammunition Testing #1/2 to 12 inches inclusive.
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provision, Federal Specification

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-636	-	Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	-	Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

SCATP-7.62MM	Ammunition Ballistic Acceptance Test Methods - Test Procedures for 7.62mm Cartridges
TECP 700-700, Vol. III	Manual of Test Methods for Small Arms Ammunition

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DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT

44021	-	Target, Incendiary
7643674	-	Classification of Cartridge Case Defects
7670238	-	Cartridge, Caliber .50, Armor Piercing Incendiary, M8
7672003	-	Cartridge, Caliber .50, Armor Piercing Incendiary, Tracer M20
7692129	-	Incendiary Flash Standards for Small Arms Ammunition Target, Incendiary

PACKAGING

12576456	-	Packing and Marking for Box, Wirebound for Cartridge, .50 Caliber
12576457	-	Packing and Marking for Box, Ammo, M2A1 for Cartridge, .50 Caliber

INSPECTION

LI 7670238	-	Inspection Equipment List for Cartridge, Caliber .50; Armor Piercing-Incendiary, M8
LI 7672003	-	Inspection Equipment List for Cartridge, Caliber .50, Armor Piercing Incendiary Tracer, M20

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

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3. REQUIREMENTS

3.1 Cartridge. The cartridges shall comply with all requirements specified in the drawings M8-7670238, M20-7672003. All associated drawings with all requirements specified in applicable specifications and standards.

3.2 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.3 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.4).

3.4 General. The cartridge shall comply with the requirements specified on Drawing 7672003 or 7670238 as applicable.

3.5 Bullet extraction. The force required to extract the bullet from the cartridge case shall be not less than 200 pounds.

3.6 Airtightness of base closure seal APIT M20. The bullet shall not release more than one bubble of air under the application of a pressure differential of 2 pounds per square inch (psi) for 5 seconds.

3.7 Residual stress. The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.8 Waterproof. The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 7 1/2 psi for 30 seconds.

3.9 Accuracy. The average of the mean radii of all targets of the sample cartridges, fired at 600 yards, shall be not greater than 12 inches.

3.10 Action time. The action time (overall primer ignition, propellant burning and bullet-barrel time) of the cartridge shall not exceed 4 milliseconds.

3.11 Velocity. The average velocity of the sample cartridges, conditioned at 68° to 72° Fahrenheit (F), shall be 2910 feet per second (ft/sec) plus or minus 30 ft/sec at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 36 ft/sec.

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3.12 Chamber pressure. The average chamber pressure of the sample cartridges, conditioned at 68° to 72°F , shall not exceed 55,000 psi.

3.13 Trace, APIT M20. When viewed at night from a line parallel to the plane of the trajectory, with the line of sight perpendicular to the plane of the trajectory at each point of observation, the bullet of the trace cartridge shall exhibit a visible trace of full luminosity from a point not greater than 100 yards from the muzzle of the weapon to a point not less than 1600 yards from the muzzle.

3.14 Incendiary flash. The incendiary composition of bullets of the sample cartridges shall ignite and produce an incandescent flash when fired against an aluminum target (Drawing 44021) at 175 yards. The flash shall be sufficiently intense and sustained to cause combustion of flammable liquids having a flash point not less than 100°F.

3.15 Penetration.

3.15.1 API, M8. The bullet core or bullet of the cartridge shall completely perforate 7/8 inch of armor plate (MIL-S-13812) placed at 100 yards from the muzzle of the weapon.

3.15.2 APIT M20. The bullet core or bullet of the cartridge shall completely penetrate 7/8 inch of armor plate (MIL-S-13812) placed at 100 yards from the muzzle of the weapon.

3.15.3 Stripping. The jacket of the bullet or any part thereof, shall not separate from the core when the cartridge is fired.

3.16 Function and casualty. The cartridges shall function without casualty.

3.17 Workmanship. The requirements for workmanship are as specified on applicable drawings, referenced specifications and the following:

3.17.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.17.2 Foreign matter. The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.2.1. Initial production sample failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

PART DESCRIPTION	Drawing	Quantity
Ctg., Cal .50, Armor Piercing Incendiary, M8	7670238	1880
Ctg., Cal .50, Armor Piercing Incendiary, Tracer M20	7672003	1880

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 2	DRAWING NUMBER
	Cartridges, Caliber .50, Armor-Piercing Incendiary M8 and Tracer M20			7670238, 7672003
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Examination for defects	1880 100%	3.3	4.4.2.1.to 4.4.2 Critical 1-6 Major 101 to 111 Minor 201 to 222
	Cartridge 1. Misfire 2. Bullet remaining in bore	1880 acc-0 rej-1	3.3	
	3. Primer defects: a. Perforation in firing pin indent in primer cup	acc-23 rej-24	3.3	
	b. Escape of gas through primer cup other than 3a	acc-6 rej-7	3.3	
	c. Escape of gas around primer cup more than 50% of periphery	acc-13 rej-14	3.3	
	d. Blow primer or primer falls out of pocket or retraction of bolt	acc-0 rej-1		

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-C-3066C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridges, Caliber .50 Armor Piercing Incendiary M8	2 OF 2	7670238 7672003
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	4. Case Casualties: 1/			
	a. Longitudinal Split			
	(1) Neck and shoulder (I or S)	acc-26 rej-27	3.3	
	(2) Body (J)	acc-6 rej-7	3.3	
	(3) Body (K)	acc-0 rej-1	3.3	
	(4) To head (L)	acc-0 rej-1	3.3	
	(5) Through head (L)	acc-0 rej-1	3.3	
	b. Circumferential rupture			
	(1) Partial, shoulder or body (J,K or S)			
	(2) Partial, head (L)			
	5. Failure to extract	acc-0 rej-1	3.3	
	6. Weapon stoppage	acc-0 rej-1	3.3	
	7. Bullet integrity	acc-0 rej-1	3.3	

NOTES:

1/ For location of defects, indicated by letters in parentheses, see Drawing 7643674

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II . Attributes sample inspection.

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.7). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ctg, Cal .50: Armor Piercing Incendiary, M8, and Armor Piercing Incendiary Tracer, M20	1 OF 2	7670238 & 7672003
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Split case in K, L or M location	100%	3.1	Visual 1/
2.	Perforated case	100%	3.1	Visual
3.	Primer missing	100%	3.1	Visual
4.	Cocked primer	100%	3.1	Visual
5.	Inverted primer	100%	3.1	Visual
6.	Weight, min 2/	100%	3.1	Scale
MAJOR				
101.	Corroded, or stained, if etched	Level III	3.1	Visual
102.	Round head	Level III	3.1	Visual
103.	Split case in I, S or J location	Level III	3.1	Visual
104.	Beveled underside of head	Level III	3.1	Visual
105.	Case mouth not crimped in cannelure	Level III	3.1	Visual
106.	No chamfer on head (rim)	Level III	3.1	Visual
107.	No visible evidence of mouth anneal	Level III	3.1	Visual
108.	Split bullet jacket	Level III	3.1	Visual
109.	Loose bullet	Level III	3.1	Visual
110.	Missing cannelure	Level III	3.1	Visual
111.	Loose primer	Level III	3.1	Visual

NOTES:

1/ Refer to MIL-STD-636 for visual defects standards.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 100 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 100 grains of propellant shall be classed as a critical defect.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ctg, Cal .50: Armor Piercing Incendiary, M8 and Armor Piercing Incendiary Tracer, M20	2 OF 2	7670238 & 7672003
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MINOR				
201.	Discolored, dirty, oily, smeared	Level IV	3.1	Visual
202.	Dent	Level IV	3.1	Visual
203.	Draw scratch	Level IV	3.1	Visual
204.	Scratch	Level IV	3.1	Visual
205.	Scaly metal	Level IV	3.1	Visual
206.	Fold	Level IV	3.1	Visual
207.	Wrinkle	Level IV	3.1	Visual
208.	Buckle	Level IV	3.1	Visual
209.	Bulge	Level IV	3.1	Visual
210.	Illegible/missing head stamp	Level IV	3.1	Visual
211.	Defective head	Level IV	3.1	Visual
212.	Defective mouth	Level IV	3.1	Visual
213.	Dent	Level IV	3.1	Visual
214.	Scratch	Level IV	3.1	Visual
215.	Scaly metal	Level IV	3.1	Visual
216.	Upset (crooked) point	Level IV	3.1	Visual
217.	Exposed steel (clad pocket)	Level IV	3.1	Visual
218.	Blunt point	Level IV	3.1	Visual
219.	Defective cannelure	Level IV	3.1	Visual
220.	Nicked/dented primer	Level IV	3.1	Visual
221.	No waterproofing material (primer pocket joint)	Level IV	3.1	Visual
222.	Defective crimp	Level IV	3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-3066C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ctg, Cal 50: Armor Piercing Incendiary M8 and Armor Piercing Incendiary Tracer, M20	1 OF 1	7670238 & 7672003
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Total length	100%	3.1	SMTE
102.	Cartridge profile failure (requiring move than 80 lbs. dead weight to insert in profile and alignment gage)	100%	3.1	SMTE
103.	Diameter of extractor groove, max.	100%	3.1	SMTE
104.	Diameter of head	100%	3.1	SMTE
105.	Thickness of head	100%	3.1	SMTE
106.	Length to shoulder datum	100%	3.1	SMTE
107.	Depth of primer	100%	3.1	SMTE
MINOR				
201.	Diameter of extractor groove, min.	Level IV	3.1	SMTE

NOTES:

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4.4.3 Tests. The tests listed in Table III shall be conducted in accordance with the methods and procedures specified in 4.4.

4.4.3.1 Test sample. The quantities for the various tests shall be as specified in Table III. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

TABLE III. Cartridge tests.

TEST	No. CTGS	REQ PARA
Bullet Extraction 1/	25	3.5
Airtightness of Base Closure Seal 2/ (APIT, M20)	--	3.6
Residual stress (Mercurous nitrate) 1/	50	3.7
Waterproof 3/	20	3.8
Accuracy 4/	90	3.9
Action time 1/	50	3.10
Velocity 4/	20	3.11
Chamber pressure 4/	20	3.12
Trace 5/ (APIT, M20)	100	3.13
Incendiary flash 6/	20	3.14
Penetration 7/	20	3.15
Stripping 8/	--	
Function and Casualty 9/	--	3.16
Gun, Machine Cal. .50 Browning		3.16
M2, HB (Turret type)	300	3.16
Gun, Machine Cal. .50 Browning		3.16
M2, HB (Flexible type)	300	3.16
Gun, Machine Cal. .50 M85 (Fixed Type)	300	3.16
Gun, Machine Cal. .50 AN-M3 Aircraft (Basic)	300	3.16

See next page for notes.

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1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ The bullets extracted from the cartridges in the bullet extraction test shall be used for this test. Failure of seven or more bullets to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than seven bullets fail in the first test, a second sample consisting of double the number of bullets in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, seven or more bullets fail to comply with the applicable requirements.

3/ Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

4/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot, subject to testing a second sample consisting of double the quantity of cartridges used in the initial test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

5/ Failure of thirty-one or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than ten but less than thirty-one cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, thirty-one or more cartridges fail to comply with the applicable requirement.

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6/ The lot shall be rejected when upon completion of the test the films showing the flash of the bullets under test are compared with the flash standards (Drawings 7692129) are less than 70 out of a possible score of 100. No second sample is permitted.

7/ a. API, M8 - Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

b. APIT, M20 - Failure of five or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than two but less than five cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, five or more cartridges fail to comply with the applicable requirement.

8/ Determination of compliance with the bullet stripping requirement shall be made during the action time test.

9/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample in Table IV. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in all the service weapons specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

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4.4.3.2 Firing defects. Firing defects and acceptance numbers shall be as specified in Table IV.

TABLE IV. Firing defects.

Defects	Acceptance First Sample	Acceptance Cumulative (1st/2nd Sample)
1. Misfire Excluding missing vent hole	1	2
2. Misfire-Venthole missing or blocked	0	1
3. Bullet remaining in bore 1/	0	-
4. Primer defects		
a. Perforation in firing pin indent in primer cup	24	60
b. Escape of gas through primer cup other than 3a.	7	15
c. Escape of gas around primer cup more than 50% of periphery	14	30
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
5. Case casualties		
a. Longitudinal split 2/		
(1) Neck and shoulder (I or S)	27	65
(2) Body (J)	7	15
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, shoulder or body (J & S)	1	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
6. Failure to extract	0	1
7. Weapon stoppage 3/	0	1
8. Bullet bursting before striking terminal or target 4/	0	-

See next page for Notes.

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1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parentheses, see Drawing 7643674.

3/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

4/ This defect shall be counted in the incendiary flash test only.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection.

4.5.1 Bullet extraction. The cartridges shall be tested in accordance with SCATP-7.62MM. The rate of travel of the test head shall be not less than three or more than six inches per minute.

4.5.2 Airtightness of base closure seal, APIT, M20. The bullets shall be placed base up under approximately 1.5 inch head of freshly boiled water in a vacuum jar test unit. The desiccator or jar shall be evacuated to a pressure of 2 pounds per square inch (4 inches of mercury) below atmospheric pressure and held at that pressure for not less than 5 seconds. The number of bubbles liberated from each bullet shall be observed.

4.5.3 Residual stress (Mercurous Nitrate). The test shall be conducted in accordance with SCATP-7.62MM.

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4.5.4 Waterproof. The cartridges (not exceeding five at one time) shall be placed horizontally under a 2 to 2 1/2 inch head of freshly boiled water in a glass container and the container shall be evacuated to a pressure of 7 1/2 pounds per square inch (15 inches for mercury) below atmospheric pressure and held at that pressure for 30 seconds. The number of bubbles liberated from the mouth or primer or both of each cartridge shall be observed.

4.5.5 Accuracy. the test shall be conducted in accordance with TECP 700-700, Vol. III.

4.5.6 Action time. The test shall be conducted in accordance with methods and procedures SCATP-7.62MM, using equipment listed in LI 7670238 or LI 7672003, as applicable.

4.5.7 Velocity. The test shall be conducted in accordance with TECP 700-700, vol. III.

4.5.8 Chamber pressure. The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.5.9 Trace (APIT, M20). The test shall be conducted in accordance with methods and procedures of TECP 700-700, Vol. III and the following:

Observation for trace performance shall be conducted on a range of at least 1600 yards in length. All trace testing shall be performed at night. Observation for trace performance shall be made at the weapon and at points 100 yards and 1600 yards beyond the muzzle of the gun on a line parallel to and approximately 75 yards from the plane of trajectory. Lights or other suitable visible markers shall be placed opposite the 100 and 1600 yards observing points to insure a line of sight from each perpendicular to the plane of the trajectory. The ammunition shall not be conditioned to temperature but shall be fired under existing conditions. Three cartridges shall be fired to sight, warm and foul each weapon prior to the test. The sample cartridges shall be fired in regular sequence permitting sufficient time between shots for each observer to record trace characteristics at each observation point. Each cartridge shall be considered as a defective one only in the event it fails at more than one observation point. During the performance of this test, observation shall also be made for (a) bullet bursting before striking terminal or target; (b) trace muzzle flash and (c) erratic flight.

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4.5.10 Incendiary flash. The test shall be conducted in accordance with TECP 700-700, vol. III.

4.5.11 Penetration. The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.5.12 Stripping. The test shall be conducted simultaneously with the action time test. The size of the paper target shall be 6' x 6'; this size may be achieved by stapling two pieces of paper together. One paper screens (6' x 6') shall be placed 50 feet from the muzzle of the gun, so that the bullets will pass through the screen near its center. The screen (s) shall be recovered or replaced as necessary to facilitate observation. At completion of the test, each target screen shall be examined, in detail at close range, for any irregular perforations which provide evidence of stripping. Any evidence of stripping shall be classified as a defect.

4.5.13 Function and casualty. In these firings, the weapons shall be at room temperature at the beginning of the test, and the machine guns shall be cooled between bursts. The number of cartridges to be fired shall be as specified in Table III. The test shall be conducted in accordance with TECP 700-700, Vol. III and as indicated below:

a. Gun, Machine, Caliber .50, Browning, M2 HB (Turret Type) - Fire in bursts of 100 cartridges.

b. Bun, Machine, Caliber .50, Browning, M2 HB (Flexible Type) - Fire in bursts of 100 cartridges.

c. Gun, Machine, Caliber .50 Tank, M85 (Fixed) - Fire in bursts of 100 cartridges.

d. Gun, Machine, Caliber .50, AN-M3, Aircraft (Basic) - Fire in bursts of 100 cartridges.

4.5.14 Defect penalty. In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table IV prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table IV.

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4.5.15 Packaging, packing and marking inspection. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy or blank cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

5. PACKAGING

5.1 Packaging (Level A). Packaging for the cartridge shall be in accordance with drawing 12576457.

5.2 Packing (Level A). Packing for the cartridge shall be in accordance with drawing 12576456.

5.3 Marking. Packaging and packing shall be in accordance with drawing 12576457 and 12576456.

5.4 General.

5.4.1 Inspection for packaging, packing and marking. Inspection for packaging, packing and marking shall be in accordance with 4.5.15.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridges are intended for use in .50 caliber machine guns.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

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e. Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

f. Requirements for contractor to provide and maintain and inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAC-C, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the US ARDEC, one (1) copy of all ballistic data and the ammunition data card for each lot shall be forwarded to: Commander, ARDEC, ATTN: AMSTA-AR-QAC-C, Picatinny Arsenal, NJ 07806-5000.

6.5 Hazard notice. The cartridge described herein and some of its components are flammable and explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to prevent fire, explosion, adverse environment, rough incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.7 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

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6.8 Subject term (keyword) listing.
Small arms
Machine gun ammunition

6.9 Changes from previous issue. Marginal notation are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-0E94)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-3066 (AR)	950123

3. DOCUMENT TITLE CARTRIDGES, CALIBER .50: ARMOR-PIERCING-INCENDIARY, M8; AND ARMOR-PIERCING-INCENDIARY-TRACER, M20

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: AMSTA-AR-EDE-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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